

SANJEEVANI.AI-Every Second Saves a Life

1. Introduction

Sanjeevani.AI is a smart ambulance routing system that helps emergency vehicles reach hospitals quickly and safely. In critical situations, traffic, road construction, and accidents can cause delays. This system uses Artificial Intelligence to find the safest and fastest route by analyzing real-time traffic and road conditions. It also sends automatic SMS alerts to traffic control authorities to clear signals for the ambulance. Sanjeevani.AI aims to reduce response time and save more lives through intelligent route planning and smart communication.

2 .Problem Statement

With the rapid increase in urban traffic and road construction activities, emergency ambulances often face serious delays in reaching hospitals. Traffic congestion, accident-prone areas, and poor coordination with traffic authorities can significantly affect response time, putting patient lives at risk. Existing navigation systems mainly focus on the shortest path and do not consider safety factors or emergency priority. There is no integrated system to predict road risks or automatically coordinate signal clearance for ambulances. Therefore, an intelligent and reliable AI-based routing system is needed to ensure faster, safer, and more efficient emergency transportation.

3. Technologies Used – Sanjeevani.AI

1. Backend – Python & Flask

Python is used as the main programming language to build the system logic, route optimization, and AI processing.

Flask is used as the web framework to connect the backend with the frontend and handle user requests, route calculations, and data processing.

2. Frontend – HTML, CSS, JavaScript

- HTML is used to design the structure of the web pages (forms, buttons, maps, dashboard)
- CSS is used to style the website and make it attractive and user-friendly.
- JavaScript is used to make the website interactive, handle dynamic updates, and display route results in real time.

Together, these technologies create a complete web-based smart ambulance routing system.

4. Data Set Information

The system uses road and traffic-related data to find the safest ambulance route. The dataset includes road distances, traffic levels, accident-prone areas, and construction zones. Each road segment is assigned a weight based on safety and congestion. This data helps the AI model select the fastest and safest route during emergencies.

5. Methodology

The system represents roads as a graph and assigns weights based on traffic, accidents, and construction.

It calculates a safety-based cost and uses a shortest-path algorithm to find the best route. Finally, it sends SMS alerts to traffic authorities to clear signals for faster ambulance movement.

6. Conclusion

Sanjeevani.AI is a smart ambulance routing system that helps emergency vehicles reach hospitals faster and safer. By using Artificial Intelligence and real-time traffic data, it selects the most efficient route instead of just the shortest one. The system also sends alerts to traffic authorities to reduce delays.

Overall, Sanjeevani.AI improves emergency response time and provides a reliable solution to modern traffic challenges, helping save more lives.

Done By:

1. G. LIKHITHA (23TK1A6618)
2. K. KEERTHANA (23TK1A6630)
3. B. SRAVANI (23TK1A6608)
4. K. NAGARAJU (23TK1A6629)
5. B. PRAVEEN (23TK1A6609)